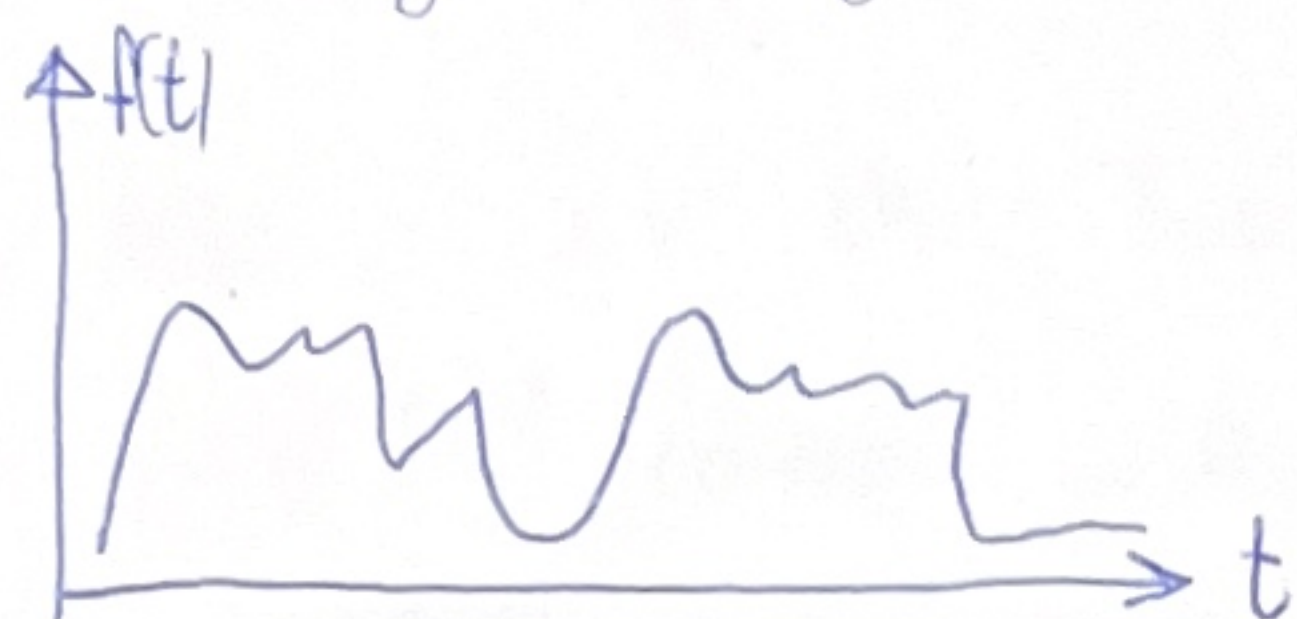


Fourier transformation

Introduction

having a signal as time series



What are the frequency components in this signal?

It is the sum of sine (and cosine) functions with different frequencies.

The basis functions

$\sin(n\omega t)$ and $\cos(n\omega t)$ $n \in \mathbb{N}$ are forming a basis of periodic functions with period $T = \frac{2\pi}{\omega}$

orthogonality:

$$\int_0^T \sin(n\omega t) \sin(m\omega t) dt = \frac{T}{2} \delta_{nm}$$

$$\int_0^T \cos(n\omega t) \cos(m\omega t) dt = \frac{T}{2} \delta_{nm}$$

$$\int_0^T \sin(n\omega t) \cos(m\omega t) dt = 0$$

The Fourier-series expansion

$f(t, T)$ can be expressed on the previous basis as

$$f(t, T) = \frac{b_0}{2} + \sum_{n=1}^{\infty} a_n \sin(n\omega t) + b_n \cos(n\omega t)$$

$$b_0 = \frac{2}{T} \int_0^T f(t, T) dt$$

$$b_n = \frac{2}{T} \int_0^T f(t, T) \cos(n\omega t) dt$$

$$a_n = \frac{2}{T} \int_0^T f(t, T) \sin(n\omega t) dt$$

The integrals can be evaluated numerically and the series expansion can be constructed

(2)

The complex formalism

for simplest numerical implementation
and to see how FFT works

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}$$

use it in the Fourier-series expansion:

$$\begin{array}{l} e^{in\omega t} \text{ comes with } c_n = \frac{a_n}{2} - \frac{b_n}{2i} \\ e^{-in\omega t} \text{ comes with } c_{-n} = \frac{a_n}{2} + \frac{b_n}{2i} \end{array} \quad \left| \quad c_0 = \frac{b_0}{2} \right.$$

So we can write:

$$f(t, T) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t}$$

$$c_n = \frac{1}{T} \int_0^T f(t) \cdot e^{-in\omega t} dt$$

Properties

- derivative

$$\mathcal{F}\left[\frac{d}{dt} f(t)\right] = i\omega \mathcal{F}[f(t)]$$

- time shift

$$\mathcal{F}[f(t-t_0)] = e^{-i\omega t_0} \mathcal{F}[f(t)]$$

Discrete Fourier Transform (DFT)

→ do the complex Fourier series expansion

on a finite number of sample points ~~$\{x_k\}$~~ $\{x_k\}$ with values $\{f_k\}$

$$\bar{f}_n = \frac{1}{N} \sum_{k=0}^{N-1} f_k e^{-2\pi i n k / N}$$

the discretized version of c_n

the inverse transform

$$f_k = \sum_{n=0}^{N-1} \bar{f}_n e^{2\pi i n k / N}$$

one can also introduce an integral form of $f(t, T)$ but
it is less important numerically

frequencies

γ_s sampling frequency

γ_{quint} : maximum resolvable freq.

$$\Delta\omega = \frac{f_s}{N}$$

$$f_k = \frac{k}{N} f_s$$

$$f_{-k} = \frac{N-k}{N} f_s$$

$$f_0 = \frac{f_s}{2}$$

(and famous)

Fast Fourier Transform

Recall:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-2\pi i k \cdot n / N}$$

kth Fourier component

nth data point

N: number of points

x_n data point

rewrite as

$$X_k = \sum_{n=0}^{N-1} x_n W_N^{nk}$$

with $W_N^{nk} = e^{-2\pi i kn / N}$

due to periodicity only $k \cdot n \pmod{N}$ matters

$O(N^2)$ multiplication, can we do better?

The $N=8$ example

$$W_8^{nk} = \begin{pmatrix} W_8^0 & W_8^0 & \dots & W_8^0 \\ W_8^0 & W_8^1 & W_8^2 & W_8^3 & W_8^4 & \dots \\ W_8^0 & W_8^2 & W_8^4 & W_8^6 & W_8^0 & \dots \\ W_8^0 & W_8^3 & W_8^6 & W_8^1 & W_8^4 & \dots \\ W_8^0 & W_8^4 & W_8^0 & W_8^4 & W_8^0 & \dots \\ W_8^0 & W_8^5 & W_8^2 & W_8^7 & W_8^1 & \dots \\ W_8^0 & W_8^6 & W_8^4 & W_8^2 & W_8^0 & \dots \\ W_8^0 & W_8^7 & W_8^6 & W_8^5 & W_8^4 & \dots \end{pmatrix}$$

Steps: /Cooley-Tukey/

- 1.) Decimation in time for x_n (split even/odd frequencies)
- 2.) Compute X_k recursively
- 3.) Use the symmetries of sine and cosine

Decimation

$$(x_0, x_1, x_2, \dots, x_7) \rightarrow (x_0, x_2, x_4, x_6), (x_1, x_3, x_5, x_7) \rightarrow (x_0, x_4), (x_2, x_6), (x_1, x_5), (x_3, x_7)$$

General decimation step

$$\begin{aligned} \sum_{n=0}^{N-1} x_n e^{-2\pi i kn / N} &= \sum_{m=0}^{N/2-1} x_{2m} e^{-2\pi i 2mk / N} + \sum_{m=0}^{N/2-1} x_{2m+1} e^{-2\pi i k(2m+1) / N} \\ &= \sum_{m=0}^{N/2-1} x_{2m} e^{-2\pi i 2mk / N} + e^{-2\pi i k / N} \sum_{m=0}^{N/2-1} x_{2m+1} e^{-2\pi i k 2m / N} \\ &= \sum_{m=0}^{N/2-1} x_{2m} W_{N/2}^{km} + W_N^k \sum_{m=0}^{N/2-1} x_{2m+1} W_{N/2}^{km} = E_k + W_N^k O_k \end{aligned}$$

and one can also prove: $X_{k+N/2} = E_k - W_N^k O_k$

- DFT separates to even/odd
- X_k and $X_{k+N/2}$ is related by symmetry
- the W_N^k factor of O_k do not depend on n and x_n

(4)

Recursive breakdown for $N=8$

$$\begin{aligned}
 X_k &= \sum_{n=0}^7 X_n W_8^{nk} = \sum_{m=0}^3 X_{2m} W_4^{km} + W_8^k \sum_{m=0}^3 X_{2m+1} W_4^{km} \\
 &= \left(\sum_{p=0}^1 X_{2p} W_2^{pk} + W_4^k \sum_{p=0}^1 X_{2p+1} W_2^{pk} \right) + W_8^k \left(\sum_{p=0}^1 X_{2p} W_2^{pk} + W_4^k \sum_{p=0}^1 X_{2p+1} W_2^{pk} \right) \\
 &= X_0 + X_4 W_2^k + W_4^k (X_2 + X_6 W_2^k) + W_8^k (X_1 + X_5 W_2^k) + W_4^k (X_3 + X_7 W_2^k)
 \end{aligned}$$

S1 - solve the innermost $\rightarrow$ S2 next gen. $\rightarrow$ S3

$$S_1^0 = (X_0 + X_4 W_2^k)$$

$$S_1^1 = (X_1 + X_5 W_2^k)$$

$$S_1^2 = (X_2 + X_6 W_2^k)$$

$$S_1^3 = (X_3 + X_7 W_2^k)$$

$$S_2^0 = S_1^0 + W_4^k S_1^1$$

$$\Rightarrow S_2^1 = S_1^2 + W_4^k S_1^3 \Rightarrow S_3^0 = S_2^0 + W_8^k S_2^1$$

$W_2^k \rightarrow 2 \text{ values} \rightarrow 8 \text{ values to store}$; $W_4^k \rightarrow 4 \text{ values} \rightarrow 8 \text{ to store}$; $W_8^k \rightarrow 8 \text{ values} \rightarrow 8 \text{ to store}$

Exploit symmetry

$$W_2^k = (-1)^k = (+1, -1, +1, -1, +1, -1, +1, -1)$$

$$W_4^k = e^{-\pi i k/4} = (1 - i, -1, i, 1, -1, i, 1, -i)$$

$$W_8^k = e^{-\pi i k/8} = (1, \frac{1}{\sqrt{2}}(1-i), -i, \frac{1}{\sqrt{2}}(-1-i), -1, \frac{1}{\sqrt{2}}(-1+i), i, \frac{1}{\sqrt{2}}(1+i))$$

This way we have $O(N \log N)$ computational time